

SOFTWARE ENGINEERING LAB 3

Architectural Justification: Telemedicine Slot Booking & Prescription Portal

Architectural Choice: Microservices Architecture

Description:

The system is structured as a set of independently deployable services: a Patient/Physician Portal UI (presentation layer), a Booking & Scheduling Service (manages doctor availability and slot reservations), a Video Consultation Service (generates and manages encrypted video room links), a Prescription Service (handles prescription authoring, digital signing, and PDF generation), and a Patient & Clinical Data Store (persists patient records, doctor profiles, and signed prescriptions). Each service owns its specific responsibility and communicates with others through well-defined interfaces, rather than sharing internal logic directly.

Two Reasons for This Choice:

Independent scaling needs: Video consultations are resource-intensive (real-time streaming, encryption overhead) and experience unpredictable peak loads compared to booking or prescription management. Microservices let the Video Consultation Service scale independently during high-demand periods (e.g., peak clinic hours) without over-provisioning the entire system.

Isolated security boundaries: Clinical and prescription data carry strict regulatory requirements (e.g., encryption at rest/in transit, license validation). Keeping the Prescription Service and Patient & Clinical Data Store as separate services allows tighter, independently-audited security controls around sensitive health data, without exposing that boundary to the booking or video components.

Security Advantage:

Because prescription authoring and clinical data storage are isolated in their own services, a compromise of the Portal UI or Booking Service cannot directly expose patient clinical records or prescription-signing credentials — access must go through the Prescription Service's own authenticated interface.

Performance Benefit:

Since booking and video consultation are decoupled, a sudden spike in slot-booking traffic (e.g., during open enrollment or flu

season) doesn't degrade the performance of active video consultations already in progress — each service can be scaled and monitored independently based on its own load pattern.

